

Agenda

Recap

One-Way ANOVA Test

F-ratio

Assumptions of ANOVA

Kruskal-Wallis Test

Shapiro Wilkin's Test

Levene-Test

Recap

Numerical - Categorical Relationship:

Z-test or T-test

(Binary Categorical)

Categorical - Categorical Relationship:

Chi-Square

Questions What test would you use for following Variable pairs

Dataset

Income vs Gender

③ Difference salaries

b/w M and F (2 Sample)

T-test or Z-test

Product vs Gender

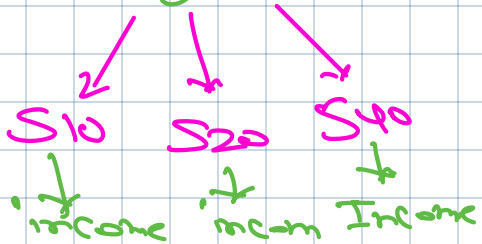
Chi-Square test of

Independence

id	Income	Gender	Product
1	10000	M	S10
2	20000	F	S20
3	18000	M	S10
4	13000	F	S40
5	19000	M	S10
...	...	...	...

Product vs
(Categorical)

Income
(Numerical)



One-Way
ANOVA Test

* ANOVA : Analysis of Variance

③ F-distribution

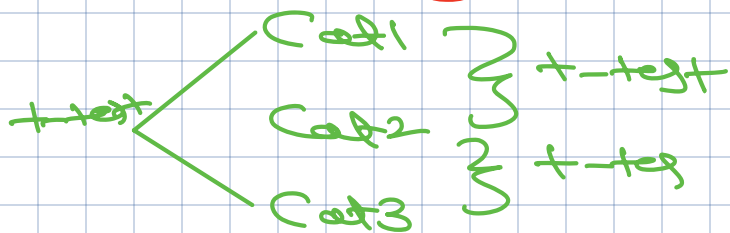
③ ANOVA is suitable for Num vs Categorical where categories ≥ 2

* Uses F-Distribution

F-statistic / F-ratio

Is there a significant difference in any of the Groups/Categories

Question: Why don't we use Z-test / T-test



Total $\rightarrow C_2$

Hypothesis for ANOVA

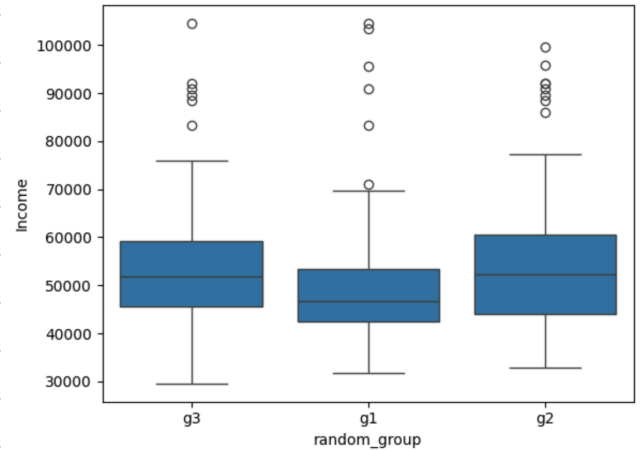
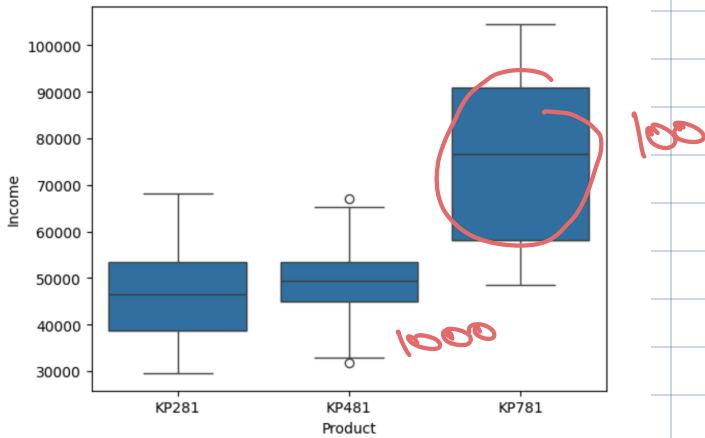
H_0 : There is no difference in mean among the Groups/Categories

H_a : There is some difference in mean of at least one of the categories

① Variance b/w Group

② Variance within Groups

Aojit Data



* $MSB \Rightarrow \sum_{i=1}^{K \text{ Group}} n_i (\bar{x}_i - \bar{\bar{x}})^2 / K$

$\bar{x}_i$ is mean of Group i

$\bar{\bar{x}}$ is Grand mean (mean of overall Data)

n_i is num-observation in i th Group

K is dof

Variance b/w Group

* $MSE \Rightarrow \sum_{i=1}^{K \text{ Group}} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 / K$

x_{ij} is j th observation of i th Group

$\bar{x}_i$ is mean of Group i

K is dof

Variance within Groups

F-ratio

F-ratio
or
F-stat

$$\frac{\text{Variance between Groups}}{\text{Variance within Groups}}$$

F-distribution

$$F\text{-ratio} \propto \frac{1}{p}$$

F-ratio is high, P-val will be low

* Interpretation

① F-score ≤ 1

P-value will be High
Fail to reject H_0

② F-score > 1

P-value will be low

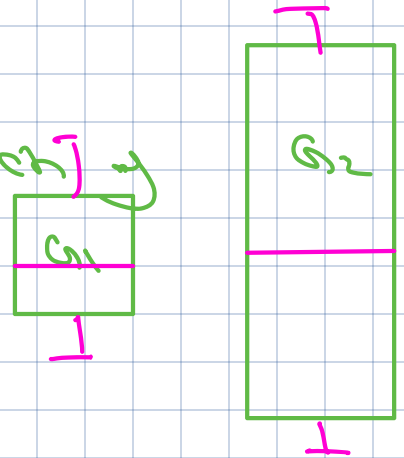
Assumptions of ANOVA

① Normality of Data

Data should be Normally Distributed

② Homogeneous Variances

Equal Variance among each of Groups



③ Independent Observations

Observations should be 'independent'.

Questions

- ① Do we have Alternative when Data is Not Gaussian?
- ② How do we check if Data is Gaussian?
- ③ How do we check for Equality of Variance?

① Do we have Alternative when Data is Not Gaussian?

Kruskal-Wallis Test

H_0 : Median of all the groups is same

H_a : At least one of the groups has significantly different median

Advantage

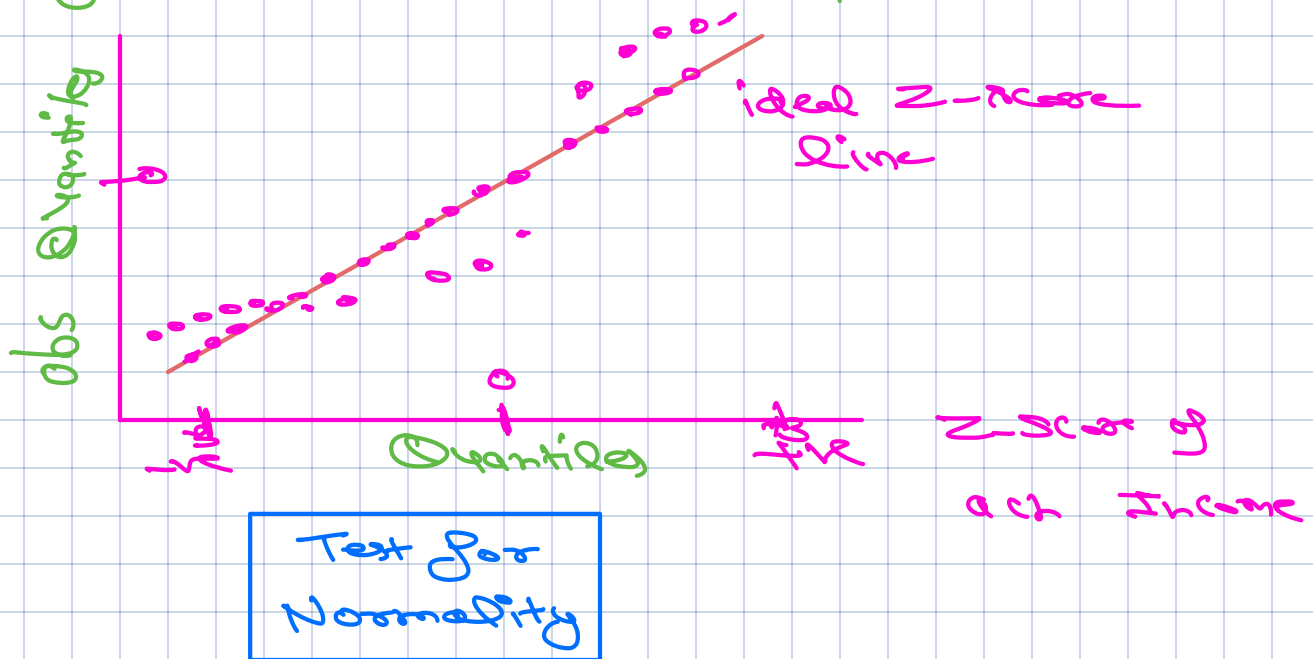
- ① Data Doesn't have to be Normal
- ② K-W test is not impacted by outliers

② How do we check if Data is Gaussian?

Ans ① Histogram

② QQ-plot

③ Hypothesis Test $\rightarrow$ Shapiro Wilk Test



Shapiro Wilkin's Test

H_0 : Data is Normally Distributed

H_a : Data is Not Normally Distributed

③ How do we check for Equality of Variance?

- ① Box plot
- ② Histogram

Levene-Test

H_0 : Variances are Equal i.e. $\sigma_1^2 = \sigma_2^2$

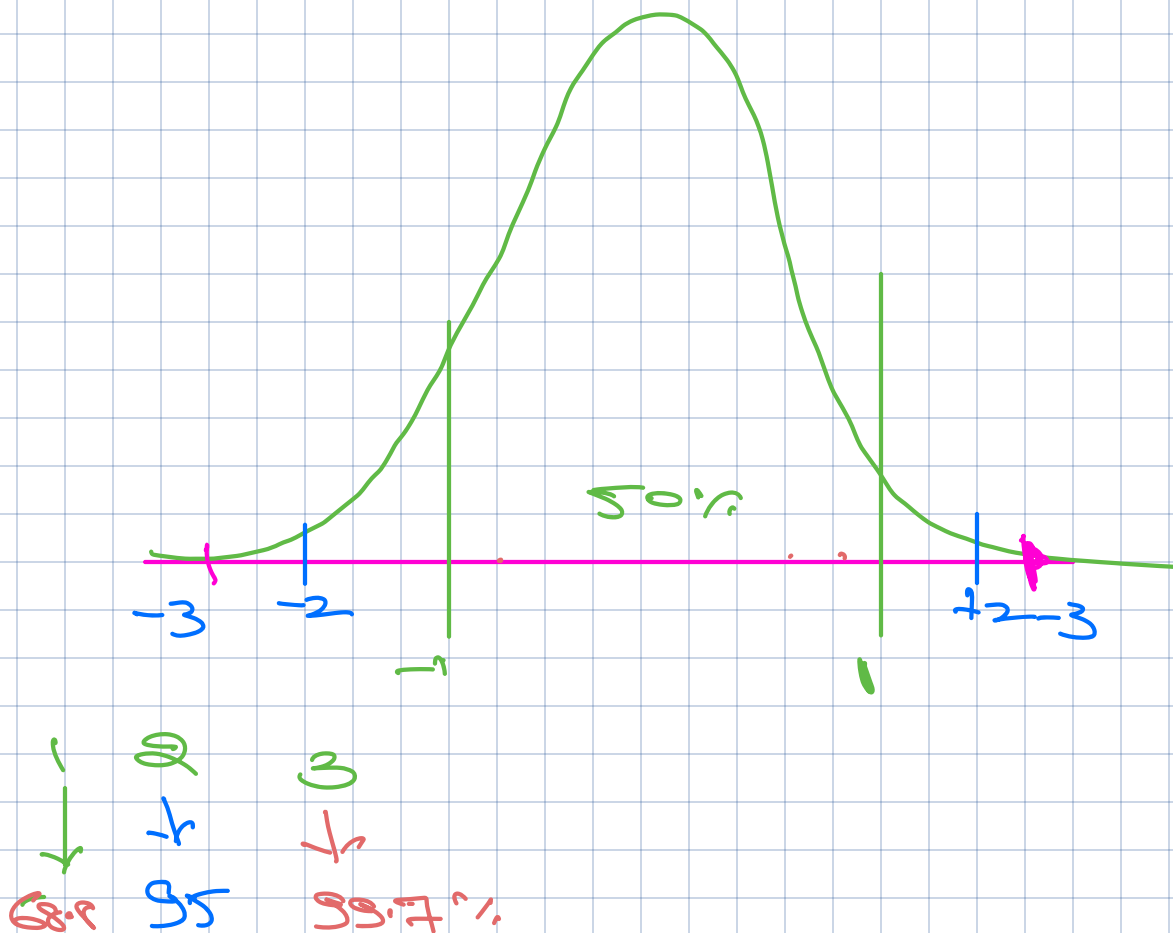
H_a : Variance are not Equal

④ One-way ANOVA

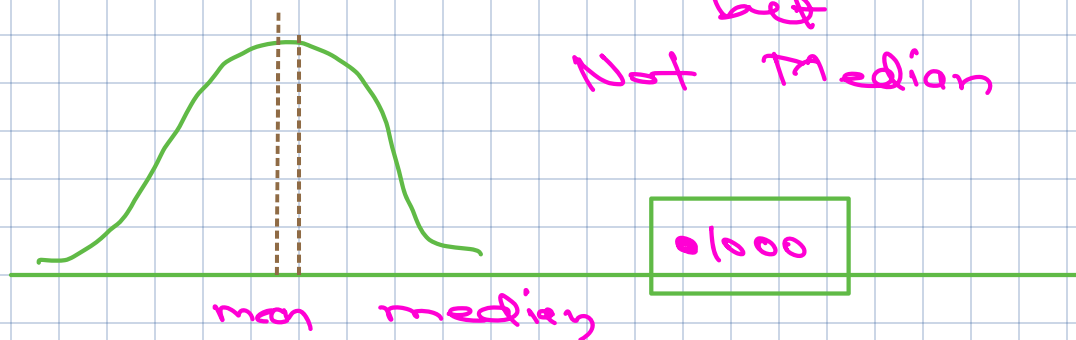
⑤ Assumption of One-way ANOVA

- ① Data must be Normally Distributed (Shapiro-Wilk)
- ② Variance of groups must be equal (Levene's Test)
- ③ Data must be Independent

⑥ QQ plot $\rightarrow$ Zscore of ideal Normal Dis
vs Actual data



How outlier impact
 Mean
 but
 Not Median



$[10, 20, 30, 40, 90] + 1000$

mean $\Rightarrow 40$

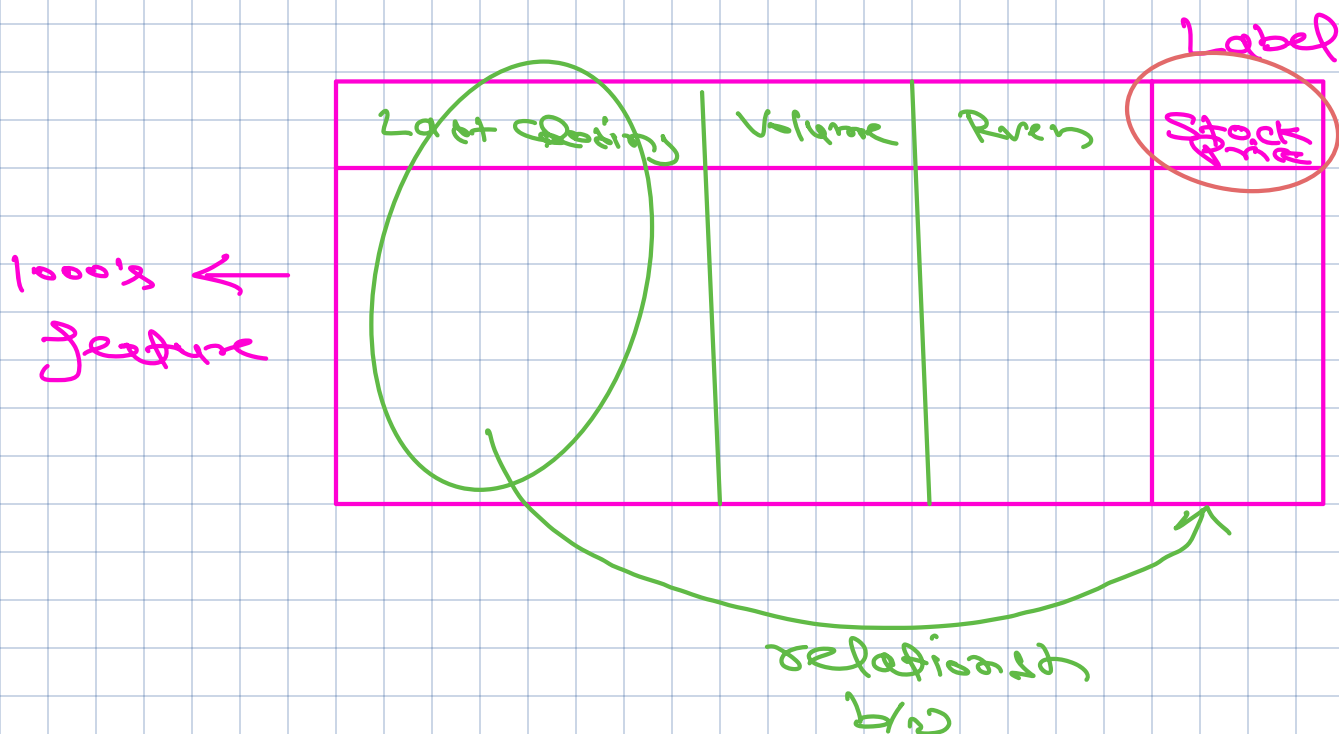
median $\Rightarrow 30$

mean $\Rightarrow > 40$

$\rightarrow$ median $\Rightarrow 30 + 40$

① Current Course

② ML Model



a feature → stock price



Paired t-test

H_0 : wt After $\geq$ wt before

H_a : wt After $<$ wt before

H_0 : wt = wt After } Two-tailed
 H_a : wt $\neq$ wt After